

ME 509: Nanomanufacturing

Course Code	ME 509
Course Name	Nanomanufacturing
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Nanoscience and Nanotechnology**
Historical developments in size reduction, fundamentals of nanoscale materials and their interactions, properties of nanocrystalline materials, size effects and quantum confinement in semiconductors, different types of nanostructures (zero, one and two dimensional) with specific examples, nanoscience in electronics, mechanics, photonics, biomedical and energy, nanomanufacturing objectives and opportunities, nanomanufacturing challenges. [6 Lectures]
- **Characterizations/fabrication techniques for nanostructures**
Basic concepts in microscopy, evolution of microscopes, electron microscopy and scanning probe microscopy for structural, microstructural, topological analysis, atomic order and compositional analysis. Application of microscopes in nanoscale characterizations. In- situ microscopy for the growth and fabrications of various nanostructures. Interface of microscopy with nanofabrication techniques. [4 Lectures]
- **Top down approaches for nanomanufacturing (subtractive)**
Concepts in top down nanomanufacturing, Mechanosynthesis-ball milling, Focused ion beam milling, thin film fabrication, thermal evaporation, E beam evaporation, Sputtering (DC, RF, reactive), thin film growth mechanism and stress evolution, Essentials of photolithography, Ebeam lithography, nanoimprint lithography, Etching methods for fabrication, dry etching and wet etching. [6 Lectures]
- **Bottom up approaches for nanomanufacturing (additive)**
Solution synthesis of nanostructures, basics of size and shape control, growth by aggregation and oriented attachment, growth from vapor phase, Atomic layer deposition, Chemical vapor deposition, Growth of carbon nanotubes, graphene and 2D materials, Vapor-liquid-solid method, Vapor phase epitaxy, Molecular beam epitaxy, Growth of important semiconductor materials; Si, GaN nanowires. [8 Lectures]
- **Advanced nanomanufacturing techniques and assembly**
Non lithographic techniques for nanomanufacturing, Template assisted methods, Template less nanopatterning, self-assembly, electric field assisted assembly. [4 Lectures]
- **Selected Topics in Nanomanufacturing and Microfabrication**
Introduction to VLSI technology, Electrochemical Processing and MEMS/NEMS technology, Challenges and Applications of nanomanufacturing in electronics, display, nanomedicine, green energy building and smart surfaces. Issues of yield and rate of production. High rate and scalable nanomanufacturing; roll to roll manufacturing for nanomaterials. Industrial R& D activities, economics and environmental concerns. [7 Lectures]

Text Books

1. Ahemed Busnaina, **Nanomanufacturing Handbook**, CRC press, 2006
2. Marc J.Madou, **Fundamentals of Microfabrication and Nanotechnology**, CRC Press, 2011
3. Ahmed & M.J Jackson, **Emerging nanotechnologies for manufacturing**, Waqar William Andrew Publishing, 2009
4. Open course materials (MIT & University of Michigan) and Journal articles